Maintenance Manual — Brookhaven Dairy Processing Facility

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Plant	Brookhaven Dairy Processing Facility	**Effective Date**	15 Jan 2026
Document Owner	B. Schroeder, Maintenance Superintendent	**Classification**	Controlled — Site Use

1. Purpose & Scope

This manual covers preventive maintenance of process equipment, ammonia refrigeration, CIP systems, and packaging lines at Brookhaven Dairy Processing Facility. Maintenance activities are coordinated with the Food Safety Plan (HACCP) to avoid creating cross-contact or contamination risks.

2. Critical Asset Register

The table below lists the critical assets covered by this manual. Each asset has a detailed task package in the CMMS (SAP-PM / Maximo) linked to its asset ID.

Asset ID	Description	Maintenance Strategy
PAS-01	HTST pasteurizer (Plate)	Plate integrity check semi-annually; FDV (flow diversion valve) test daily.
SEP-02	Cream separator	Bearing inspection every 1500 hours; bowl integrity at 6 months.
REF-NH3-01	Ammonia compressor (screw)	Vibration and oil analysis monthly; relief valve test per IIAR 6.
CIP-04	CIP skid (caustic/acid)	Quarterly verification of chemical concentration, conductivity probe calibration.
FIL-PKG-08	Gable-top filler	Daily nozzle inspection; weekly seal jaw alignment.

3. PM Schedule Summary

Frequency	Activity
Daily	FDV test on PAS-01 before each production run.
Weekly	Filler seal alignment; ammonia engine room walkdown per IIAR 6.
Monthly	Vibration on REF-NH3-01; lubricant analysis.
Quarterly	CIP probe calibration; integrity check on dairy plate heat exchangers.
Annual	Pressure relief valve test; pasteurizer holding tube re-validation.

4. Work Management

All maintenance work is authorized through a CMMS work order. Work orders are categorized as PM, PdM, Corrective, Emergency, or Project. Maintenance KPIs (Schedule Compliance ≥85%, PM Compliance ≥95%, MTBF) are reported weekly to the Plant Manager.

5. Procedures

5.1 Sanitary Maintenance

All maintenance in product zones uses dedicated, color-coded tools. Tools entering product contact surfaces are sanitized with peracetic acid prior to and after use.

5.2 FDV Validation

The Flow Diversion Valve is tested daily and after any maintenance. Any FDV failure halts pasteurizer operation and triggers a deviation under HACCP-BDP-CCP1.

5.3 Ammonia Mechanical Integrity

Performed under the IIAR 6 framework. Engine room operators document compressor parameters every 2 hours. Predictive analytics flow into the Maximo CMMS with thresholds reviewed quarterly.

6. References

- ANSI/IIAR 6 — Mechanical Integrity of Ammonia Refrigeration
- 3-A Sanitary Standards
- FDA Pasteurized Milk Ordinance
- Internal: HACCP-BDP-CCP1, MNT-BDP-PM-12

7. Revision History

Rev	Date	Author	Summary
1	04 Jun 2023	B. Schroeder	Initial issue.
2	21 Nov 2024	B. Schroeder	Added predictive analytics integration.
3	09 Jul 2025	B. Schroeder	Updated asset register.
4	15 Jan 2026	B. Schroeder	Refreshed PM frequencies and KPIs.

Approvals

Role	Name	Signature	Date
Maintenance Engineering	B. Schroeder, Maintenance Superintendent	____________________	15 Jan 2026
Reliability Engineering	Sr. Reliability Engineer	____________________	15 Jan 2026
Operations / Site Lead	C. Larsson, Plant Manager	____________________	15 Jan 2026